PR #90 · PAYPAL ON MASTER

How to test the PayPal feature

A buyer pays for AI document work through PayPal sandbox checkout, gets a receipt tied to that payment, and accepts or challenges each output. The merchant then gives free rework or refunds only the challenged part. Every step is recorded, and nothing here touches the Colosseum app.

Watch the whole cycle first (67 seconds): paypal-demo.mp4. It plays inline on the pull request too. The demo uses the real PayPal workspace screens with PayPal and the AI simulated, because the hosted service isn't deployed yet. Your test below uses the real PayPal sandbox.

The cycle at a glance

1. Quote	Buyer adds documents, gets a price. Nothing charged.
2. Pay	Buyer approves on PayPal sandbox, then confirms the capture.
3. Deliver	Paid work runs. Each claim links to a source excerpt.
4. Review	Buyer accepts or challenges each output, with a reason.
5. Resolve	Merchant approves free rework or a refund for that output.
6. Record	Receipt, dashboard and CSV agree. A redacted link can be shared.

Where it runs

One separate Vercel project, **chainpay-paypal**, at **https://chainpay-paypal.vercel.app**. The site and the API share that address. It uses its own Neon database and the PayPal sandbox only; the code refuses live PayPal. The Colosseum relay, MCP server and website are not changed.

BEFORE YOU START

Setup checklist

Full values for every setting are in **deploy/paypal/README.md** → "Vercel release". Tick these off:

#	Do this	You should see
1	Merge PR #90, then GitHub → Actions → **Deploy to Vercel** → Run workflow.	The **paypal** job goes green (it is skipped until the PAYPAL_VERCEL_PROJECT_ID variable exists).
2	Open **https://chainpay-paypal.vercel.app/healthz**.	`{"status":"ready"}`. "disabled" means CHAINPAY_PAYPAL_ENABLED is not true; "unavailable" means the database is not reachable.
3	In Vercel, set **CHAINPAY_PAYPAL_NEW_PURCHASES=true** and redeploy.	Without it the checkout button stays off. Set it back to false when you finish.
4	Check the PayPal webhook in the sandbox app points at **/v1/paypal/webhooks** and its ID is in PAYPAL_WEBHOOK_ID.	Refunds and payment changes are confirmed through this webhook.
5	Have the sandbox **personal** account email and password (developer.paypal.com → Testing tools → Sandbox accounts).	This is the account that "pays" at checkout.
6	Issue two access codes in the same workspace (below).	Two 64-character codes. Treat them like passwords.

Issue the access codes from a terminal that has the Vercel environment variables loaded:

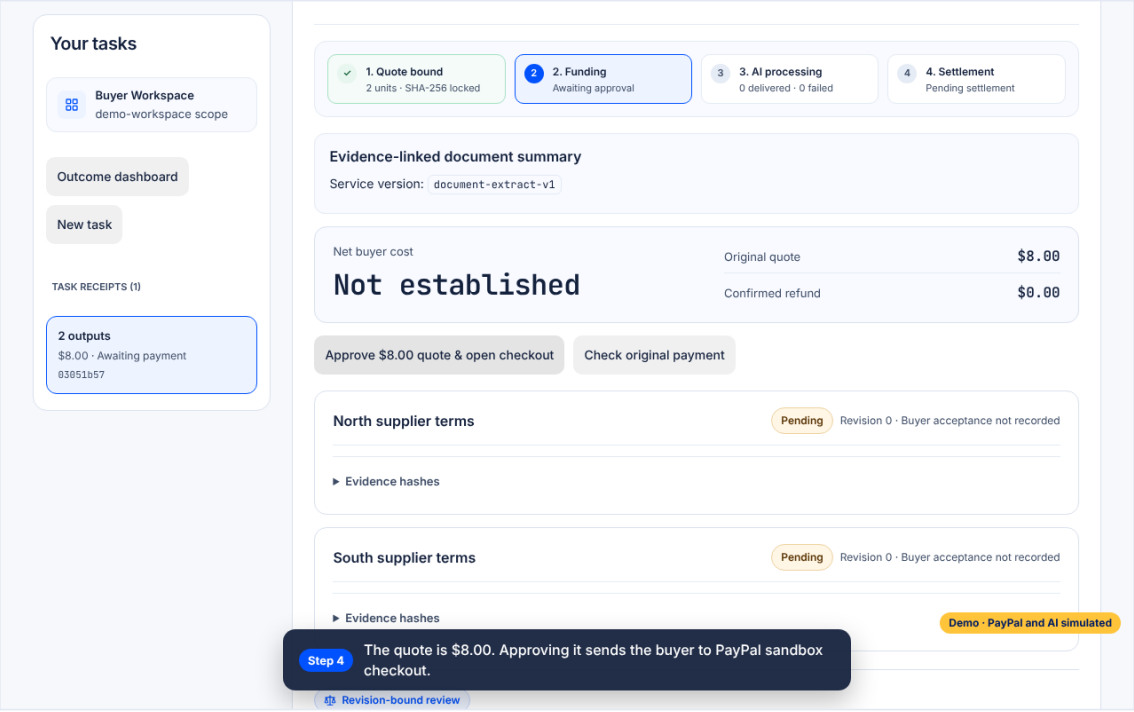
```
cargo run -p chainpay-backend --features paypal --bin chainpay-paypal -- \
  issue-access test-ws test-buyer buyer
cargo run -p chainpay-backend --features paypal --bin chainpay-paypal -- \
  issue-access test-ws test-merchant merchant
```

The paid AI step spends from the OpenRouter key in CHAINPAY_PAYPAL_AI_KEY. Make sure that key has a spend cap before you start.

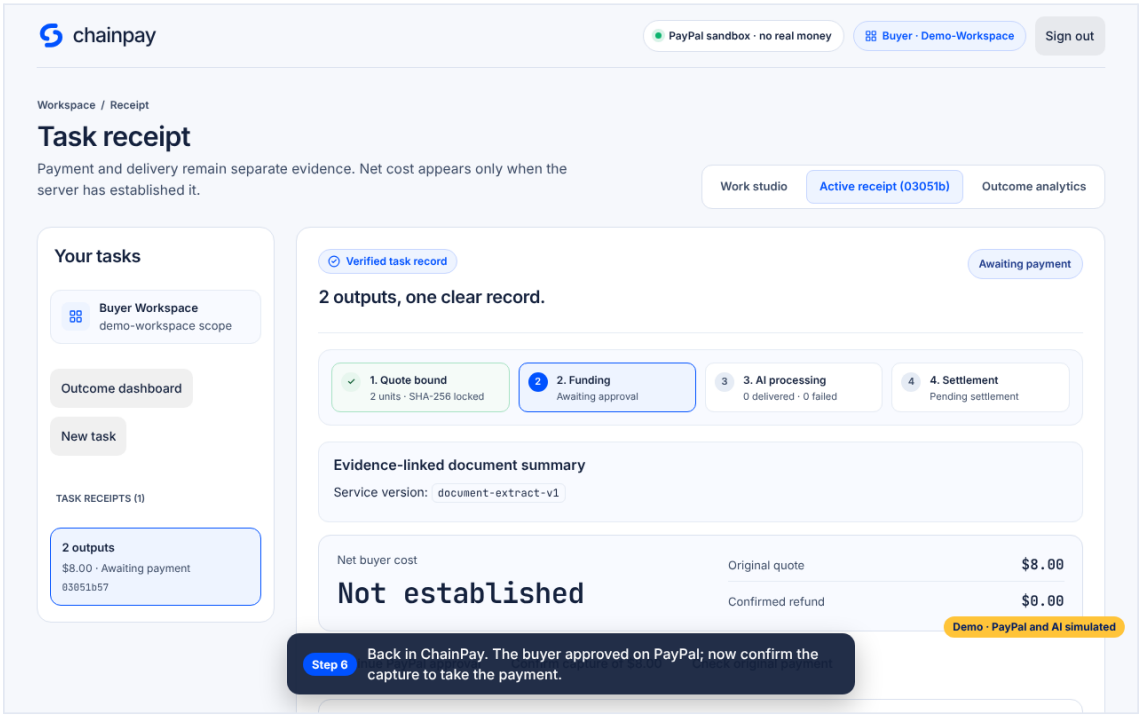
TEST 1

Buy and deliver (buyer)

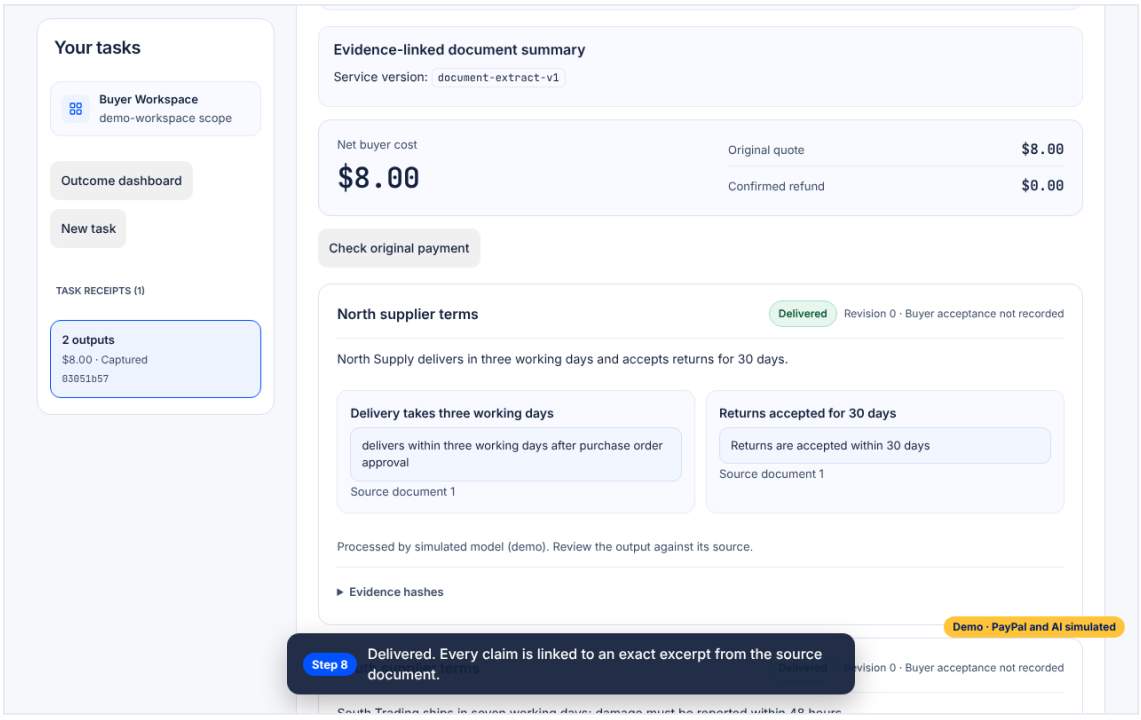
#	Do this	You should see
1	Open the site and paste the **buyer** access code.	"Prepare a quote" with the service list.
2	Choose **Evidence-linked document summary**. Paste text into Document 1, **Add document**, paste into Document 2.	Both documents marked "Readable input".
3	**Prepare quote**.	A receipt with "Awaiting payment" and a button **Approve $8.00 quote & open checkout**. Net buyer cost: Not established.
4	**Approve $8.00 quote & open checkout**.	PayPal sandbox checkout for $8.00 USD.
5	Log in with the sandbox personal account and approve.	You return to ChainPay with **Confirm capture of $8.00**.
6	**Confirm capture of $8.00**.	"PayPal capture confirmed". Net buyer cost $8.00.
7	**Run or recover this paid task**. This can take a minute.	Both outputs "Delivered", each claim with its source excerpt.



Demo step 4. The quote, before any money moves.



Demo step 6. Back from PayPal: the payment is approved but not yet captured.



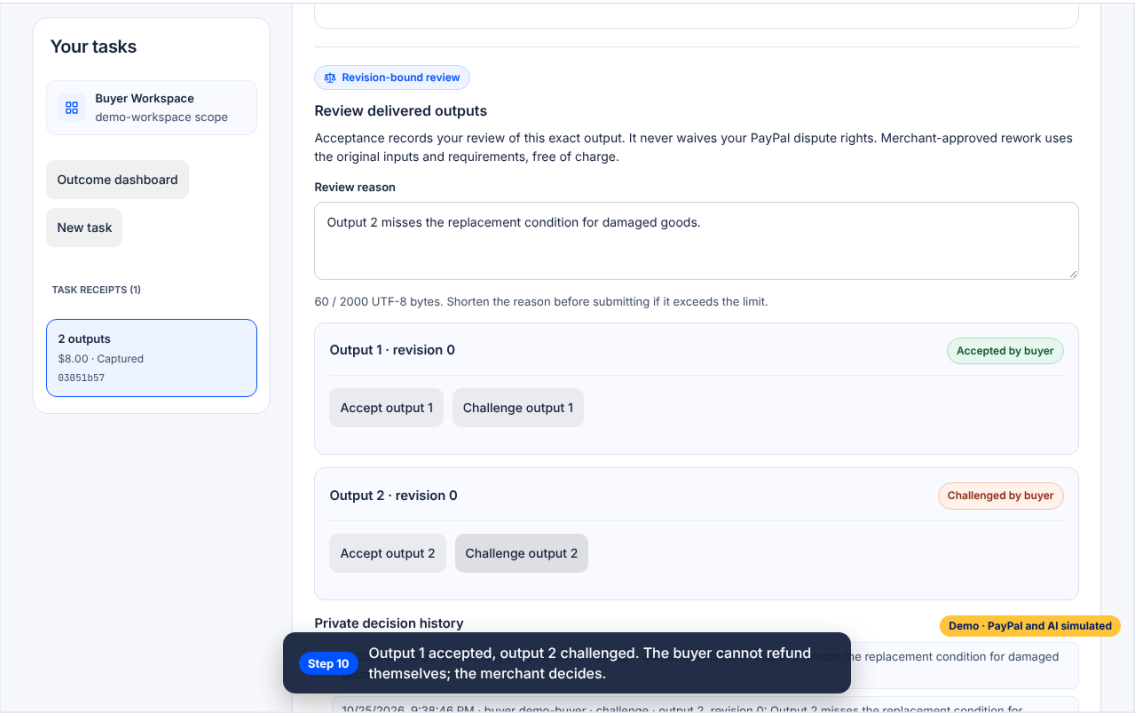
Demo step 8. Delivered output. Every claim links to the exact words in the source.

TEST 2

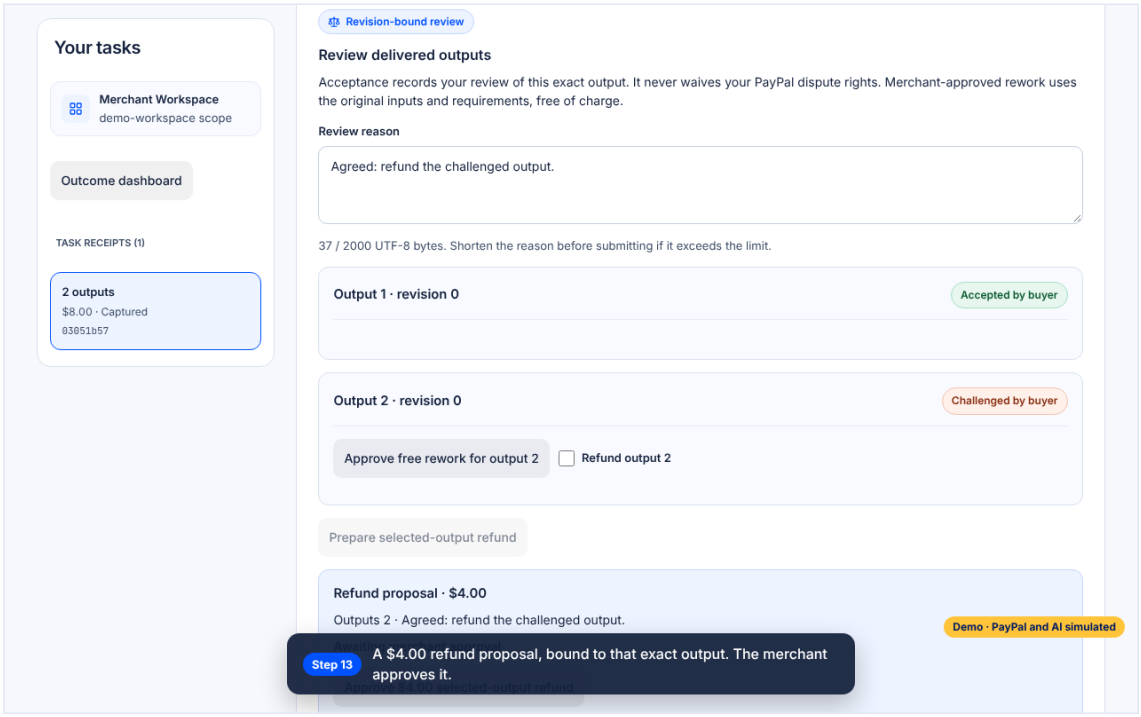
Challenge and refund (buyer, then merchant)

#	Do this	You should see
1	Still as buyer: type a **Review reason**, **Accept output 1**, **Challenge output 2**.	"Accepted by buyer" and "Challenged by buyer". The buyer has no refund button.
2	**Sign out**, paste the **merchant** code, open the same task from the list.	The same task with the buyer's decisions.
3	Type a reason, tick **Refund output 2**, **Prepare selected-output refund**.	"Refund proposal · $4.00", bound to output 2.
4	**Approve $4.00 selected-output refund**.	Within a few seconds: **Refund confirmed**, net buyer cost **$4.00**.
5	Optional, on a fresh task: **Approve free rework for output N** instead.	That output reruns at no charge as revision 1.

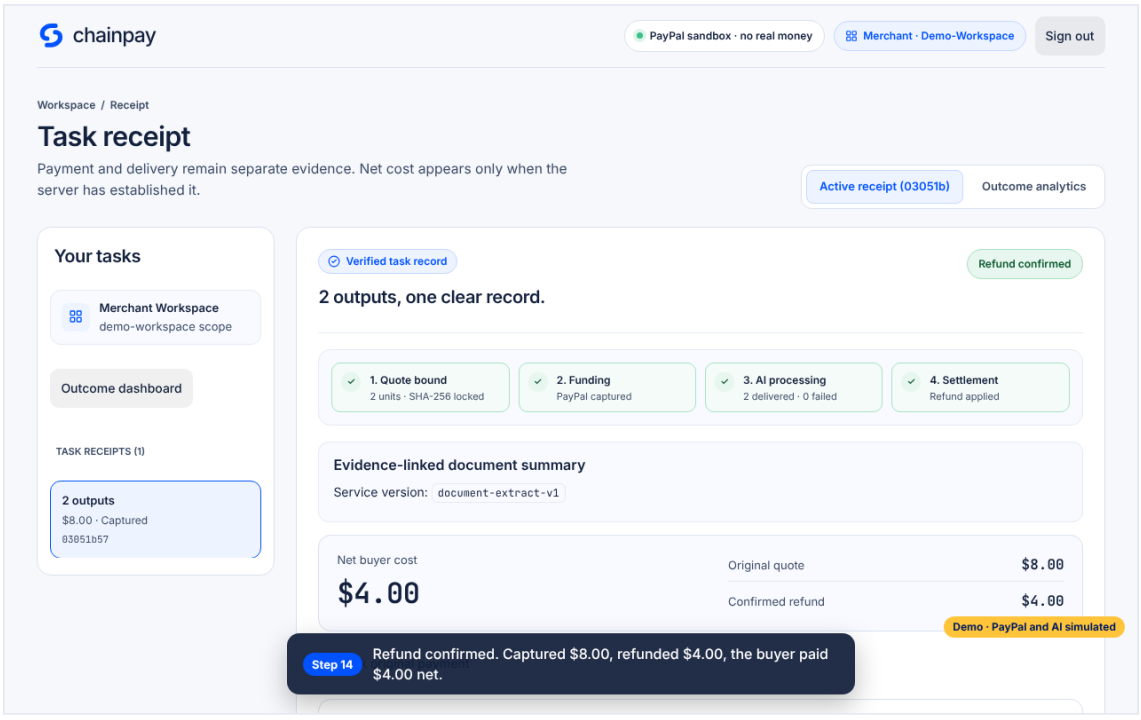
The key check in this test: after the refund the receipt must say "Refund confirmed" and show net $4.00. It must **not** say "Merchant review required". That was a bug fixed in this PR.



Demo step 10. Buyer decisions recorded per output.



Demo step 13. The merchant's refund proposal for the challenged output.

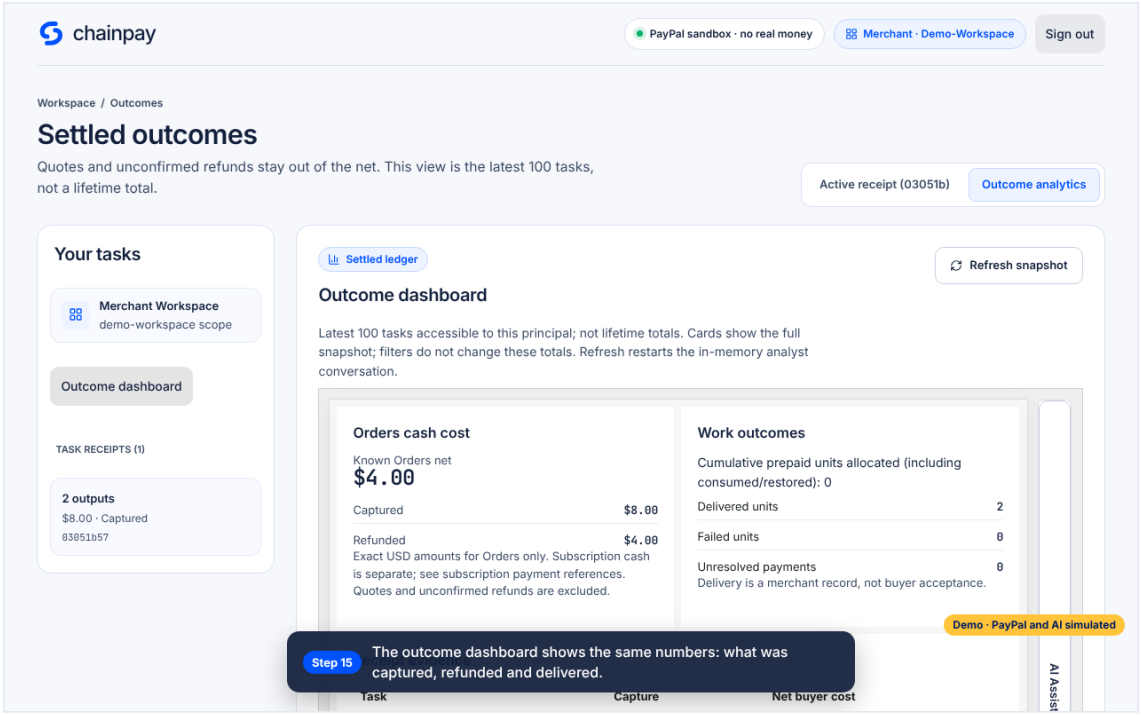


Demo step 14. After the refund: captured $8.00, refunded $4.00, net $4.00.

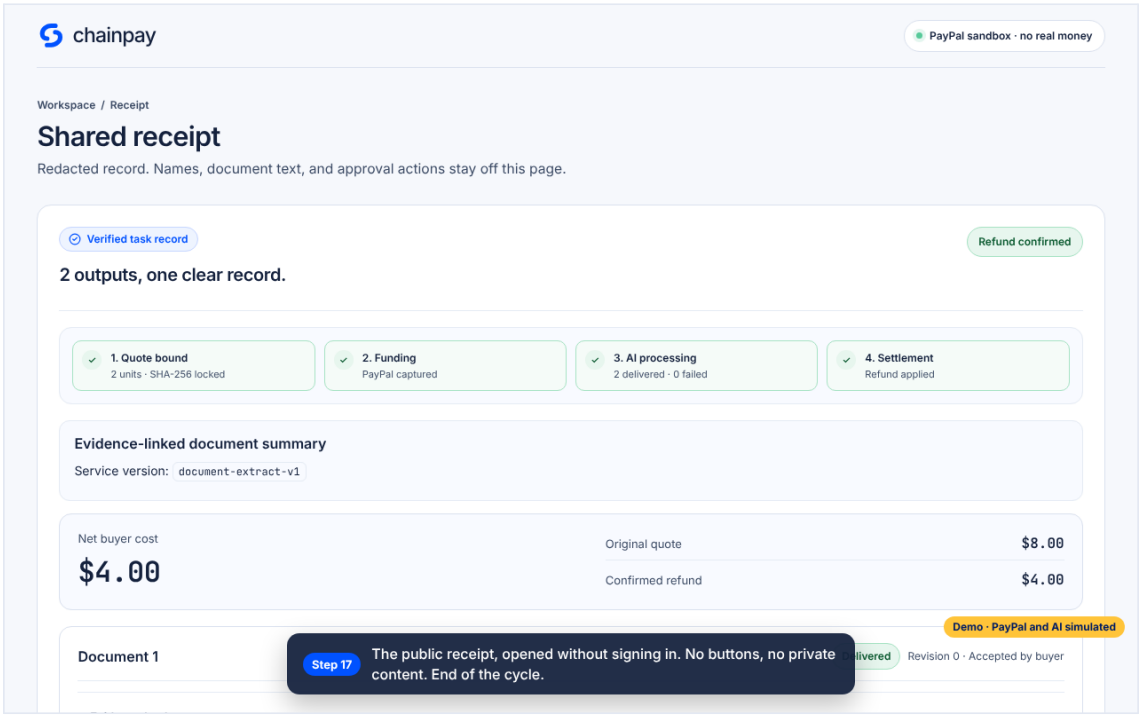
TEST 3

Records and sharing

#	Do this	You should see
1	**Outcome dashboard**.	Captured $8.00, refunded $4.00, known net $4.00, 2 delivered units.
2	**Export CSV**.	One row with the same numbers.
3	As buyer: **Share redacted receipt**, open the link in a private window.	Amounts and status only: no document text, no names, no buttons.
4	**Revoke share link**, reload the private window.	The link no longer opens.
5	Log in at sandbox.paypal.com with the sandbox **business** account.	The $8.00 payment and $4.00 refund appear there too.



Demo step 15. The outcome dashboard agrees with the receipt.



Demo step 17. The shared receipt, opened without signing in.

TEST 4

Safety checks (two minutes)

#	Do this	You should see
1	On a paid task, **Check original payment**.	It re-reads PayPal and nothing changes.
2	Click **Run or recover this paid task** again.	No second charge, delivered outputs are not redone.
3	Try the buyer access code on the merchant-only buttons (they should not show).	Buyers never see refund approval; merchants never see checkout.
4	GitHub → Actions → **PayPal inbox pass** → Run workflow.	Green, with `"complete":true`.

If something breaks

Write down the **task ID** (Provider references on the receipt) and the time. **Do not create a new task** to work around it: every payment step can be recovered from the original task, and that recovery is part of what we are testing. Use **Check original payment** first. Server logs are in Vercel → chainpay-paypal → Logs.

Checkout button missing	CHAINPAY_PAYPAL_NEW_PURCHASES is false, or the quote expired (prepare a new one).
"No response yet. Keep this task"	The request timed out but may have worked. Click Check original payment; don't retry with a new task.
Refund stays pending	The webhook has not arrived. Check the webhook URL and ID, or run the PayPal inbox pass workflow.
/healthz says disabled	CHAINPAY_PAYPAL_ENABLED is not exactly true. Redeploy after changing it.
Sign-in rejected	Access code expired (7 days by default) or from another database. Issue a new one.

Optional: subscriptions

Only if CHAINPAY_PAYPAL_SUBSCRIPTIONS=true and a $40/month plan ID is set. As buyer, **Approve $40.00 monthly subscription terms**, approve on PayPal, and the allowance shows 10 units that a task can use instead of a card checkout.